

5 The photographs show a Northwestern crow and an American crow.



(Source: © All Canada Photos / Alamy Stock Photo)



(Source: © MichaelGrantBirds / Alamy Stock Photo)

These two crows look very similar and are therefore difficult to distinguish as two separate species.

Scientists have studied the nuclear DNA and the mitochondrial DNA (mtDNA) of these two species of birds.

(a) Give **two** differences between the structure of nuclear DNA and mtDNA.

(2)

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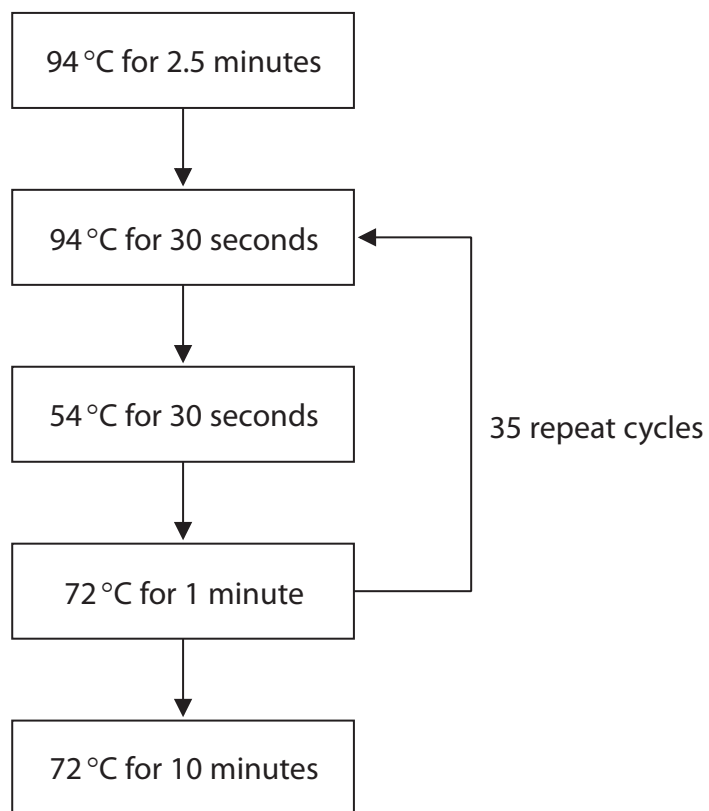
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- (b) The mtDNA was isolated from these two species of crow and amplified using the polymerase chain reaction (PCR).

The diagram shows details of the process used.



- (i) Name **three** molecules, other than the mtDNA and water, that would be needed in this process.

(2)

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- (ii) Calculate the total length of time, in hours, that this process would take.

Give your answer to two decimal places.

(2)

Answer hours

(iii) Explain how this process amplifies the DNA.

Use information in the diagram to support your answer.

(3)

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(c) Explain how the amplified mtDNA could be used to determine the genetic relationships between these two species of crow.

(3)

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(Total for Question 5 = 12 marks)